

Characterization of ashes from a 100 kW_{th} pilot-scale circulating fluidized bed with oxy-fuel combustion

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ABSTRACT

Oxy-fuel combustion experiments have been carried out on an oxygen-fired 100 kW_{th} mini-circulating fluidized bed combustion (CFBC) facility. Coal and petroleum coke were used as fuel together with different limestones (and fixed Ca:S molar ratios) premixed with the fuel, for *in situ* SO₂ capture. The bed ash (BA) and fly ash (FA) samples produced from this unit were collected and characterized to obtain physical and chemical properties of the ash samples. The characterization methods used included X-ray fluorescence (XRF), X-ray diffraction (XRD), char carbon and free lime analysis, thermogravimetric analysis (TGA), and surface analysis. The main purpose of this work is to characterize the CFBC ashes from oxy-fuel firing to obtain a better understanding of the combustion process, and to identify any significant differences from the ash generated by a conventional air-fired CFBC. The primary difference in the sulfur capture mechanism between atmospheric air-fired and oxy-fuel FBC, at typical FBC temperatures (~850 °C), is that, in the air-fired case the limestone sorbents calcine, whereas the partial pressure of CO₂ in oxy-fuel FBC is high enough to prevent calcination, and hence the sulfation process should mimic that seen in pressurized FBC (PFBC). Here, the char carbon content in the fly ash was much higher than that in the bed ash, and was also high by comparison with ash obtained from conventional commercial air-firing CFBC units. In addition, measurements of the free lime content in the bed and fly ash showed that the unreacted Ca sorbent was present primarily as CaCO₃, indicating that sulfur capture in the oxy-fuel combustor occurred *via* direct sulfation. Limestone utilization for oxy-fuel combustion in this unit was generally lower than that in industrial-scale air-firing CFBCs, with better limestone performance found during combustion of petcoke running at relatively higher temperatures. The Brunauer–Emmett–Teller (BET) surface area and also the pore volume in the fly ash were much higher than in the bed ash and smaller size pores predominated in the fly ash samples.

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1. Introduction

Carbon dioxide is one of the major greenhouse gases that contribute to anthropogenic global climate change [1] and oxy-fuel combustion is seen as one of the promising new technologies that can reduce such emissions to the atmosphere [2]. This process uses nearly pure oxygen (90%+) instead of air for combustion. To control the combustion temperature and make up the volume of missing N₂, most of the flue gas is recycled to the combustor. The CO₂ concentration is highly enriched (>90%) by recycling the flue gas and easy recovery can be achieved for further utilization or sequestration [3,4]. An additional advantage of this process is that the NO_x emissions are substantially reduced compared to the air-firing scenario since N₂ in the oxidant has been removed prior to combustion.

For fluidized bed combustion (FBC) boilers, in addition to their excellent fuel flexibility, SO₂ emissions can also be controlled *in situ* by adding limestone sorbents with the fuel (for both air-fired and oxy-fuel combustion).

Most research and development (R&D) work on oxy-fuel combustion was focused on pulverized coal combustion previously, including the earlier work at CanmetENERGY (e.g., [4,5]), and the origins can be traced to pioneering research carried out by Argonne National Laboratories in the 1980s [6]. Subsequently, in the 1990s, significant oxy-fuel R&D started elsewhere, with a number of small pilot plant programs, including CanmetENERGY (Canada), Air Liquide (US) and the International Flame Research Foundation (IFRF) R&D program in Ijmuiden (The Netherlands), looking initially at natural gas firing [3]. In addition to the early small-scale pilot plant work there were various economic evaluations of the technology *versus* back-end scrubbing, primarily for natural gas-fired systems and there are now two major reviews in the open literature available

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for interested readers, which describe such developments [3,7]. Extensive research has been carried out on emissions from oxy-fuel pulverized coal firing, such as NO_x , SO_x , as well as other micro-pollutants (e.g., [8,9]). There are also a few publications discussing ash formation under oxy-combustion of pulverized coals (e.g., [10–13]). There is already one large demonstration plant (30 MW_{th}) operating in Europe with more being planned in the future [14–16].

By contrast, to date oxy-combustion technology has received relatively little attention for oxy-fuel circulating fluidized bed combustion (CFBC), although the concept was examined over 30 years ago for bubbling FBC [17]. However, more recently oxy-fired FBC has become increasingly important as a potential technology, offering as it does fuel flexibility and the possibility of firing local or indigenous fuels, including biomass in a CO_2 -neutral or even negative manner. One example of this is a paper on a 95 kW_{th} oxy-fuel combustion bubbling FBC reported by Romeo and co-workers [18], who studied the combustion characteristics, heat transfer and pollutants emissions under different O_2/CO_2 concentrations, achieved by mixing gases from cylinders.

Currently the R&D work on oxy-fired CFBC technology is being undertaken in numerous countries, including Canada, Finland, Poland, China and the United States among others. Alstom and Foster Wheeler have explored the oxy-fuel CFBC concept using pilot-scale tests [19,20]. Alstom's work included tests in a unit of up to 3 MW_{th} in size, but did not involve recycle of flue gas [21]. Foster Wheeler's work [19] also involved pilot-scale testing, using a small (30–100 kW) CFBC owned and operated by VTT (Technical Research Centre of Finland) and this work along with CanmetENERGY's work with its own 100 kW CFB combustor [22–24] appears to be the first in which units were operated with oxy-fuel combustion FBC using flue gas recycle. CanmetENERGY has two pilot plants which are capable of being operated in the oxy-fuel mode, with full flue gas recycle: a nominally 75 kW unit and a larger 0.8 MW_{th} unit. Foster Wheeler has recently completed 8 months of trials at CanmetENERGY using the larger 0.8 MW_{th} unit [25], as a prelude to building a 30 MW_{th} oxy-fuel CFBC demonstration plant in Spain [26]. Foster Wheeler is also the first company to commercialize supercritical CFBC technology (Łagisza Power Plant, Poland) and with this technology as the basis, it is now working with the power company ENDESA (Spain) on the technology development of a 300 MW_e supercritical Flexi-burn™ CFBC, which would allow a CFBC boiler to operate either in air- or oxy-firing conditions. The predicted CO_2 capture for the Flexi-burn CFBC technology is 90% of emissions and it is anticipated that it could be available by 2020 [27].

Other recent work has been reported by Czakiert and co-workers [28], who studied fuel conversion (i.e., sulfur, nitrogen and carbon) in a 100 kW_{th} oxygen-enriched air combustion CFB (oxygen concentration up to 28%) with no flue gas recycle. They concluded that combustion efficiency improved with elevated oxygen partial pressure and conversion ratio of fuel- N_2 to NO_x increased under oxy-firing conditions. Krzywanski et al. [29] have developed a mathematical model for oxy-fuel CFBC which considered hydrodynamics of bed material, fuel combustion, flue gas desulfurization, and heat and mass transfer. Again, the major research interest has been focused on emissions of SO_x , NO_x and CO produced from oxy-fuel FBC combustion. Although there are a few publications regarding oxy-fuel pulverized coal combustion ash, no open literature can be found so far discussing the properties of oxy-fuel FBC ash. It should be noted that Wang et al. [30] discussed the influence of the water content in flue gas on carbonation of fly ash under oxy-fuel CFB conditions. They suggested the carbonation effects are likely to be enhanced by water over a wide range of

temperatures. However, it should also be noted that the fly ash used was produced from an air-fired CFB boiler and the focus of their study was to investigate the potential for back-end fouling due to calcined limestone carbonation, for situations where indirect sulfation occurs in the bed under oxy-fuel combustion conditions. Namely, they were concerned with the combustion of unreactive fuels, which would require high temperatures of 900 °C and above, as a result of which indirect sulfation would occur in the bed, after which fouling due to carbonation of the CaO in the spent sorbent might arise at temperatures of 800 °C or less, in flue gases with a high CO_2 partial pressures as is the case with an oxy-fuel CFBC.

In the current work, a series of ash samples produced from CanmetENERGY's 100 kW mini-CFBC reactor during oxy-fuel combustion of different feedstocks was collected and analyzed. Various chemical and physical analysis methods, including X-ray fluorescence (XRF), X-ray diffraction (XRD), char carbon and free lime analysis, thermogravimetric analysis (TGA), and surface analysis, were utilized to characterize these ashes. The aim of this work was to provide the first published information on ashes produced from oxy-fuel combustion in CFBC. This will allow a better understanding of both the process and ash disposal issues, as well as allowing a comparison of oxy-fuel combustion ash with conventional air-firing CFBC ash. In the configuration of oxy-fuel FBC with flue gas recirculation, the partial pressure of CO_2 is high enough to prevent calcination of limestone particles. The sulfur capture mechanism under this condition, which should be similar to that occurring in the pressurized FBC (PFBC), is also discussed. It should be noted that since there are significant differences existing between oxy-fuel pulverized coal and FBC combustion, in terms of the operating temperature range, particle size range, and addition of limestone for SO_2 capture in FBC, the comparison of ashes produced from these two different systems is not investigated in this paper.

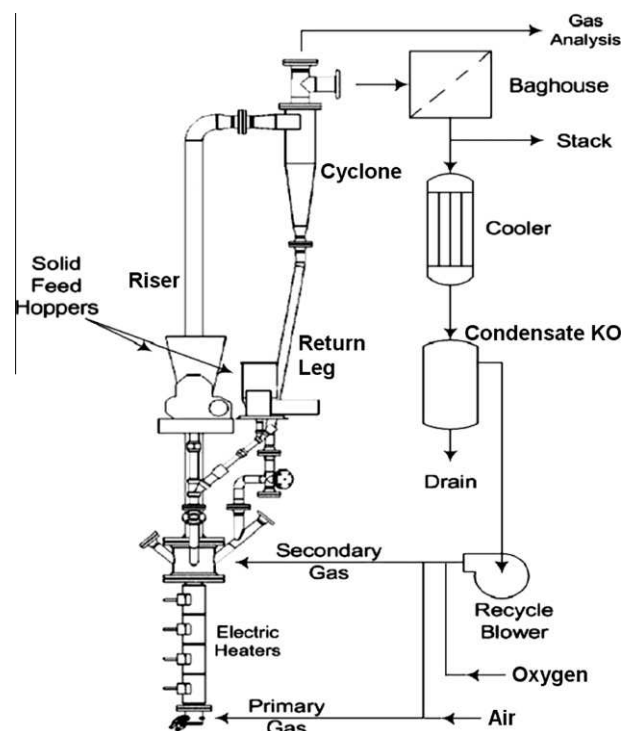


Fig. 1. Schematic of the mini-CFBC at CanmetENERGY.

2. Methods/equipment

2.1. Mini-CFBC

A schematic of the 100 kW_{th} mini-CFBC facility is given in Fig. 1. The mini-CFBC consists of a riser, cyclone, return leg and fuel/sorbent feeder. The refractory-lined riser has an internal diameter of 100 mm and internal height of 5 m. It is equipped with both ash removal and flue gas analysis systems. Locations for ash sample collection include the bottom of the baghouse, the return leg and the fluidized bed. Flue gas is continuously monitored for NO_x, SO₂, CO, CO₂ and O₂. Solids pass from the combustor through a transition section into a cyclone and down through the return leg to re-enter the combustor directly above the bottom bed section. Four stock ceramic-insulated band heaters (18 kW in total) surround the bottom section of the combustor to help achieve appropriate combustion temperatures quickly. These heaters are computer controlled and, if necessary, can be run during operation to help maintain isothermal conditions in the dense bed zone. Solid feed rates of up to 15 kg/h can be employed and the bed temperature maintained in the range of 750–950 °C, while the superficial gas velocity can be run at up to 8 m/s. The mini-CFBC has been extensively modified for oxy-fuel CFBC combustion, including the addition of oxygen supply line and the flue gas recycle train, i.e., the recycle blower, etc. The details of typical operating procedures have been described elsewhere [23,24].

Prior to performing oxy-fuel combustion experiments, the mini-CFBC was pre-heated with the external electric heaters, and fuels were gradually fed in the air-firing mode when the bed temperature was above the ignition temperature. When the bed temperature reached the desired level under air-firing conditions, the air supply was shut off simultaneously with the flue gas recycle blower being switched on. Oxygen was supplied to the mini-CFBC with the recycled flue gas to replace the combustion air. The tran-

sition from air-firing to oxy-fuel-firing mode was smooth and easily controlled, and typically took only a few minutes. The mini-CFBC was operated at a nominal bed temperature of 850 °C. Superficial gas velocity was about 4 m/s. Global oxygen concentration during oxy-fuel combustion periods was ~28–30%. The nominal Ca:S molar ratio used here was 2–3 [23], which is typical of atmospheric FBC systems.

2.2. Fuel, limestone and bed material

The ashes collected and analyzed in this work were obtained during experiments in which two coals, namely Eastern bituminous coal (EB, a low-sulfur coal from US) and East Kentucky coal (K, also a bituminous coal, with medium-sulfur content), and one petroleum coke (C, high-sulfur), were fired. Fuels were crushed to <5 mm before combustion. The fuel and fuel ash analysis results are presented in Tables 1 and 2. The two coals contain 8.8% and 12% of ash, respectively, in which SiO₂ and Al₂O₃ are the main components. The petroleum coke is a low-ash fuel (1%). However, there are significant amounts of vanadium and nickel in the coke ash, which is not seen in the coal ashes but is normal for petroleum coke. SiO₂ and Al₂O₃ are the other major components in the coke ash.

Two different limestones were used for *in situ* sulfur capture in the mini-CFBC. Havelock (HL) is a high-purity Canadian limestone (from New Brunswick) with a size range of 0–0.5 mm. Czatkowice (CZ) limestone is from Poland with a size range of 0–0.8 mm. Table 3 gives the characteristics of the two limestones. Limestone was premixed with the fuel and fed into the CFBC through the fuel feeder. Sand (typically in a batch of about 5 kg) with a size range of 0.15–0.35 mm was loaded as an initial bed material in the pre-heat stage. Silica sand (>70% SiO₂, provided by KPM Company) was used in most of the tests. However, olivine synthetic sand (produced by Olimag Sands Inc.), which contains considerable amounts of Mg (~25%) and SiO₂ (~40%), was used in the last two experiments (tests 10 and 11).

2.3. Ash collection and characterization

Several oxy-fuel combustion experiments were conducted on this facility using different combinations of fuel and limestone. Bed ash (BA) and fly ash (FA) samples were collected after each of these tests for further characterization. Table 4 summarizes the conditions under which the ash samples were produced. A Ca/S molar ratio of 3 was used in most tests, except for limited runs with EB coal and petcoke, which used a Ca/S molar ratio of 2 (with HL limestone). FA2T and FA2B represent the top and the bottom portion of the fly ash captured in the baghouse ash bin. In some cases return leg ash (RL) and bed drain ash (BD) were also collected during the experiment. BD7 was the ash drained from the bottom of the fluidized bed during steady state under oxy-fuel combustion with the bed temperature at about 850 °C. To allow for experimental error, repetitions were conducted for fuels with addition of HL limestone. Multiple ash samples for the same fuel and limestone combination in different experiments were therefore collected.

Table 1
Analysis of fuels, wt.%.

	Eastern bituminous coal (EB)	East Kentucky coal (K)	Petroleum coke (C)
<i>Proximate analysis (dry)</i>			
Moisture (as analyzed)	1.08	2.38	0.66
Ash	8.86	12.27	1.0
Volatile matter	35.78	35.79	11.46
Fixed carbon	55.56	51.94	87.54
<i>Ultimate analysis (dry)</i>			
Carbon	77.81	73.17	86.91
Hydrogen	5.05	5.01	3.22
Nitrogen	1.49	1.62	1.83
Sulfur	0.95	1.52	5.88
Ash	8.86	12.27	1.0
Oxygen (by difference)	6.04	6.41	1.16
Heating value (MJ/kg)	32.51	30.01	34.71

Table 2
Fuel ash analysis, wt.%.

Major component	Eastern bituminous coal (EB)	East Kentucky coal (K)	Petroleum coke (C)
SiO ₂	56.38	49.71	40.13
Al ₂ O ₃	29.27	29.80	13.90
Fe ₂ O ₃	6.55	12.09	5.65
CaO	1.2	0.96	4.36
MgO	0.86	0.99	0.8
SO ₃	1.03	0.86	0.61
V ₂ O ₅	Trace	Trace	27.0
NiO	Trace	Trace	4.199

Table 3
Analysis of limestones, wt.%.

Composition	Havelock (HL)	Czatkowice (CZ)
CaCO ₃	96.4	97
MgO	0.59	0.48
SiO ₂	1.23	1.9 ^a

^a Including Fe₂O₃ + Al₂O₃ + Na₂O + K₂O + S.

Table 4
Ash production conditions.

Ash samples	Fuel	Limestone	Ca/ S	Bed temperature
BA1, FA1	Eastern	Havelock	2	840–900 °C
BA10, FA10	Bituminous	Havelock	3	
BA2, FA2T, FA2B	East Kentucky	Havelock	3	820–920 °C
BA3, FA3			3	
BA8, FA8			3	
BA4, FA4	Petcoke	Havelock	2	840–960 °C
BA5, FA5, RL5			3	
BA6, FA6			3	
BA7, FA7, BD7(850)	Petcoke	Czatkowice (0.1– 0.8 mm)	3	840–950 °C
BA9, FA9	East Kentucky	Czatkowice (0.1– 0.8 mm)	3	840–900 °C
BA11, FA11	Eastern bituminous	Czatkowice (0– 0.8 mm)	3	820–900 °C

Table 5
XRF and char carbon analysis of ashes from Eastern Bituminous coal + limestone combustion, wt.%.

Major component	Eastern Bituminous + Havelock		Eastern Bituminous + Czatkowice	
	BA (EB + HL)	FA (EB + HL)	BA11	FA11
SiO ₂	37.5 ± 0.03	24.2 ± 0.4	31.21	22.61
Al ₂ O ₃	6.86 ± 5.3	13.1 ± 1.1	3.58	13.36
Fe ₂ O ₃	4.11 ± 3.6	3.15 ± 1.1	6.23	2.37
CaO	21.6 ± 11.5	8.79 ± 7.0	22.8	7.31
MgO	0.6 (BA1)	0.3 (FA1)	23.88	1.11
	30.5 (BA10)	2.75 (FA10)		
SO ₃	5.98 ± 4.3	4.08 ± 2.1	4.41	3.77
Loss on fusion	6.09 ± 2.9	42.9 ± 11.2	6.69	47.47
Sum	100.0	100.	100	100
Char carbon (by TGA)	0.28 ± 0.20	36.54 ± 8.75	0.49	40.34
Limestone utilization	18.0 ± 4.2	37.5 ± 12.0	14	36

The collected ash samples were subjected to various analyses, including XRF, XRD, char carbon and free lime analysis, TGA and surface analysis, to characterize physical and chemical properties. Free lime is defined as the sum of CaO + Ca(OH)₂ expressed as CaO percent. This content was examined using the sucrose method as described in ASTM C-25. Details of this method can be found elsewhere [31]. The ash samples were pyrolyzed in a TGA (Mettler Toledo TGA/SDTA851^e/LF/1100 °C) at up to 900 °C in either a N₂ or air environment to determine char carbon and CaCO₃ content. The surface analysis was conducted with a Micromeritics ASAP 2000 instrument using nitrogen physisorption technique at 77 K.

Table 6
XRF and char carbon analysis of ashes from East Kentucky coal + limestone combustion, wt.%.

Major component	East Kentucky + Havelock				East Kentucky + Czatkowice	
	BA(K + HL)	FA(K + HL)	FA2T	FA2B	BA9	FA9
SiO ₂	39.6 ± 12.8	18.1 ± 2.2	18.89	20.39	34.49	20.96
Al ₂ O ₃	11.9 ± 2.4	10.1 ± 1.0	10.94	10.8	9.29	11.65
Fe ₂ O ₃	3.64 ± 1.36	4.51 ± 1.16	5.39	2.86	4.29	5.57
CaO	26.2 ± 7.5	19.6 ± 7.5	18.79	30.51	32.7	12.46
MgO	1.86 ± 0.20	1.58 ± 0.27	1.82	1.79	1.14	0.87
SO ₃	8.25 ± 3.58	8.19 ± 1.70	6.69	8.78	8.1	7.34
Loss on fusion	5.13 ± 6.0	35.7 ± 9.1	35.15	23.0	6.74	38.74
Sum	99.9 ± 0.2	99.6 ± 0.3	99.64	99.61	100	100
Char carbon (by TGA)	0.14 ± 0.11	23.0 ± 7.7	23.14	12.13	0.59	31.22
Limestone utilization	21.4 ± 3.5	31.9 ± 12.5	25	20	17	41

3. Results and discussion

3.1. XRF and char carbon analysis

Tables 5–7 present XRF and char carbon analysis results of BA and FA produced from different fuels used in this work with limestone addition under oxy-firing conditions. In these tables, loss on fusion refers to the sample weight loss at 1100 °C, tested according to ASTM D 4326; limestone utilization is defined as the number of moles of CaO reacted with SO₂ divided by the initial number of moles of CaO. Details of the char carbon content determination are described later in the TGA analysis section.

The typical char carbon content in bed ash or fly ash produced from commercial-scale CFBC (conventional air-firing) ranges from 0.5% to 3% [31]. In all cases fly ash obtained from this unit had much more char carbon (>20%) than in the bed ash (typically less than 1.25%, which is in reasonable agreement with commercial-scale CFBC ash). The high char carbon content in the fly ash is believed to be mainly due to the unavoidably short residence time in small-scale CFBC test facilities such as the one used here.

In coal-firing cases (as shown in Tables 5 and 6), results for BA and FA followed similar patterns for both limestones. Limestone utilization in coal-firing fly ash, with either HL or CZ limestone as sorbent, was 1.5–2.5 times as high as that in the bed ash. In the case of petcoke-firing (as shown in Table 7), when HL limestone was used, utilization in both BA and FA was at a relatively consistent level of ~30%. When CZ limestone was used in oxy-firing with petcoke, limestone utilization in FA was again much higher than in BA (43% vs. 27%), which is similar to results found in the coal-firing cases. In coal combustion, the two limestones performed basically the same, while both limestones performed generally better for petcoke combustion, which operated at a slightly higher temperature of ~950 °C (Table 4). These results were in line with the *in situ* measurement reported by Jia et al. [24].

In Table 6, FA2T and FA2B represent the top and the bottom portion of the fly ash captured in the baghouse ash bin. There were apparent discrepancies between these two ash samples in terms of char carbon content and limestone utilization, which can probably be explained because FA2B was captured at the beginning of the oxy-fuel combustion steady state, while FA2T was collected after running at steady state for a significantly longer time (up to 24 h). Return leg ash (RL5) was also collected during coke combustion with addition of HL limestone (Table 7). This ash was more like the bed ash with very low char carbon content and relatively high limestone utilization. It should be noted that, in the ash of BD7 (bed drain ash at bed temperature of 850 °C), there was a significant amount of char carbon detected (Table 7). This was normally seen in the fly ashes but not in the bed ashes. This ash was drained from the bed at steady state of oxy-firing at about 850 °C, and it is apparent that the char carbon had insufficient time to burn com-

Table 7

XRF and char carbon analysis of ashes from Coke + limestone combustion, wt.%.

Major component	Coke + Havelock			Coke + Czatkwice		
	BA(C + HL)	FA(C + HL)	RL5	BA7	FA7	BD7(850)
SiO ₂	30.8 ± 10.7	13.1 ± 7.9	39.45	11.48	10.84	17.5
Al ₂ O ₃	5.70 ± 2.3	6.03 ± 3.7	6.49	2.59	5.47	3.02
Fe ₂ O ₃	2.36 ± 0.6	3.30 ± 2.1	1.87	1.59	2.91	0.94
CaO	34.2 ± 10.3	26.2 ± 3.7	23.83	55.71	22.08	42.27
MgO	2.31 ± 0.4	2.10 ± 0.3	1.8	1.22	1.14	0.99
SO ₃	14.8 ± 5.0	11.2 ± 1.2	9.5	21.8	13.59	11.78
Loss on fusion	6.85 ± 6.78	36.3 ± 11.3	13.4	4.09	42.12	21.85
Sum	99.8 ± 0.2	99.8 ± 0.2	99.59	99.99	100.00	99.99
Char carbon (by TGA)	0.37 ± 0.5	19.3 ± 8.0	0.91	1.25	29.59	6.09
Limestone utilization	30.4 ± 6.2	30.0 ± 1.1	28	27	43	20

pletely. Limestone utilization in this ash was at a relatively low level of 20% compared to the BA and FA from the same experiment.

Limestone utilization in the oxy-fuel combustion ash of this work (especially the coal-fired ash) was generally lower than that in typical air-fired CFBC ash, which is in the range of 30–45% for a 160 MW_e commercial-scale CFBC [31]. However, other work on the mini-CFBC has confirmed that limestone utilizations are also lower for oxy-fired combustion compared with air-firing for the same unit as well [23,24]. The sulfur capture mechanism under oxy-fuel combustion environment will be discussed below.

3.2. XRD analysis

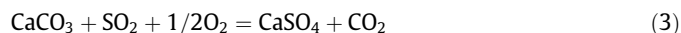
XRD analysis results of ashes produced from different fuels with limestone addition are given in Tables 8–10. In either coal- or coke-firing mode, the amount of Ca(OH)₂ was at a low level (typically <5%), except for the BA9 sample, which showed 49% portlandite, indicating that this ash might have absorbed moisture before analysis. It is noted in Table 8 that a significant amount of magnesium compounds (45–60%) was found in BA10 and BA11 samples, consistent with the XRF results shown in Table 5, because olivine sand, which contains ~25% Mg, was used in these two tests. The sand typically stayed in the dense bed section and no significant magnesium was found in the fly ash samples.

The CaCO₃ and CaSO₄ components were in the range of 10–20% for both BA and FA in the case of East Kentucky coal combustion with addition of HL limestone (Table 9). These two components analyzed lower when the same fuel was burned with addition of CZ limestone. CaCO₃ and CaSO₄ contents were even lower (<10%) in the ashes produced from EB coal combustion with addition of either limestone (Table 8).

In the case of coke plus limestone firing, the CaCO₃ and CaSO₄ contents were typically higher, as shown in Table 10. It can be seen that, in the case of coke with HL limestone, the CaCO₃ content in both bed ash and fly ash was substantially higher than that pro-

duced from coal-fired ashes. The anhydrite content in these ashes was higher than that of other ashes, indicating that in this unit sulfur capture performance was better in oxy-fuel combustion with coke. This was in accordance with XRF results and previous findings [24]. It should also be noted that there was more than 40% calcite found in the BD7 ash sample, which was drained from the bed at 850 °C. This suggests that the limestone was not sufficiently utilized due to the short residence time.

In a typical air-fired CFBC ash, calcite was normally not detectable and anhydrite was found at higher levels, e.g., 60–80% [31,32]. In the air-firing mode limestone calcined to completion very quickly after entering the combustor with the fuel and reacted with SO₂ (i.e., indirect sulfation via Eqs. (1) and (2)).



The existence of CaCO₃ in the ash supports the belief that the sulfur capture process in oxy-fuel combustion was via the direct sulfation mode (Eq. (3)) without calcination of the limestone, because the CO₂ partial pressure in the flue gas was high enough to prevent calcination from occurring. The anhydrite content in this work was found to be lower than that in air-fired CFBC ash. This fact may imply poorer limestone performance for oxy-fuel combustion compared to the air-firing mode; however, as indicated above the ashes previously examined by Wu et al. [31,32] were obtained from a much larger CFBC unit. In fact, one can find arguments in the literature that direct sulfation, which occurs in oxy-fuel combustion at typical FBC temperatures (850 °C), might promote limestone performance, because counter-diffusion of the CO₂ generated during direct sulfation (Eq. (3)) forms a porous product layer that offers less diffusion resistance than the essentially nonporous layer formed during indirect sulfation [33]. This

Table 8

XRD analysis of ashes from Eastern Bituminous coal + limestone combustion, wt.%.

	Eastern Bituminous + Havelock		Eastern Bituminous + Czatkwice	
	BA(EB + HL)	FA(EB + HL)	BA11	FA11
Quartz (SiO ₂)	3.95 ± 3.46	9.60 ± 0.99	1.3	7.1
Lime (CaO)	4.05 ± 0.49	1.85 ± 2.47	8.4	1.2
Calcite (CaCO ₃)	4.40 ± 1.41	2.95 ± 3.18	7.4	3.4
Anhydrite (CaSO ₄)	7.20 ± 7.1	10.7 ± 6.4	6.0	9.8
Portlandite (Ca(OH) ₂)	0.40	n/a	0.3	n/a
Forsterite (Mg ₂ SiO ₄)	38.5 (BA10)	n/a	27.7	n/a
Proto-enstatite (MgSiO ₃)	22.2 (BA10)	n/a	17.7	n/a
Crystallinity	62.0 ± 17.8	33.0 ± 10.1	72.1	30.7
Amorphous	38.0 ± 17.8	67.1 ± 10.1	27.9	69.3

Table 9

XRD analysis of ashes from East Kentucky coal + limestone combustion, wt.%.

	East Kentucky + Havelock				East Kentucky + Czatkowice	
	BA(K + HL)	FA(K + HL)	FA2T	FA2B	BA9	FA9
Quartz (SiO ₂)	14.8 ± 5.5	6.15 ± 0.84	6.6	7.1	9.2	8.2
Lime (CaO)	5.3 ± 0.8	2.93 ± 2.86	3.1	6.9	2.8	2.6
Calcite (CaCO ₃)	11.0 ± 13.6	16.4 ± 5.4	15.1	22.4	6.5	5.4
Anhydrite (CaSO ₄)	17.1 ± 12.2	18.2 ± 3.6	14.9	21.8	8.3	20.5
Portlandite (Ca(OH) ₂)	1.93 ± 1.88	4.75 ± 3.32	2.4	7.1	48.9	n/a
Crystallinity	86.2 ± 5.6	53.8 ± 14.3	49.8	74.9	97.8	52
Amorphous	13.8 ± 5.6	46.2 ± 14.3	50.2	25.1	2.2	48

Table 10

XRD analysis of ashes from Coke + limestone combustion, wt.%.

	Coke + Havelock			Coke + Czatkowice		
	BA(C + HL)	FA(C + HL)	RL5	BA7	FA7	BD7(850)
Quartz (SiO ₂)	12.0 ± 4.9	6.07 ± 4.72	28.6	5.7	3.0	11.4
Lime (CaO)	1.73 ± 0.6	2.33 ± 1.75	0.9	14.4	2.5	2.5
Calcite (CaCO ₃)	20.8 ± 22.6	36.5 ± 27.1	10.0	7.0	13.4	40.1
Anhydrite (CaSO ₄)	26.2 ± 8.6	30.3 ± 5.7	29.8	26.9	33.9	27.9
Portlandite (Ca(OH) ₂)	4.30 ± 1.47	3.40 ± 2.12	n/a	n/a	n/a	n/a
Crystallinity	83.5 ± 21.3	81.9 ± 29.9	90.5	70.9	55.2	108.5
Amorphous	16.5 ± 21.3	18.1 ± 29.9	9.5	29.1	44.8	−8.5

would be expected to lead to much higher limestone utilization. However, it should also be noted that actual measurements on ashes from two large PFBCs, in which the partial pressure of CO₂ was high enough (similar to that in the oxy-fuel combustion environment) to prevent calcination of limestone sorbent, failed to demonstrate such superior utilization [34,35], and it has recently been argued that the gaseous back-diffusion argument is in error. Hu et al. [36] provided evidence that the porosity in the product layer results from the formation of crystal grains and does not depend on the evolution of CO₂ during the sulfation process. Hu et al. [37] also indicated that high CO₂ concentration can significantly

impede the direct sulfation reaction, possibly by slowing the diffusion of CO₂ away from the limestone particles, which is in line with the conclusion that indirect sulfation performed better for oxy-fuel CFBC of petcoke at 950 °C [24], where indirect sulfation is expected to be the dominant sulfation mechanism at such temperature. The current work, based on the analysis of oxy-fuel combustion ash, supports the findings of Jia et al. [24].

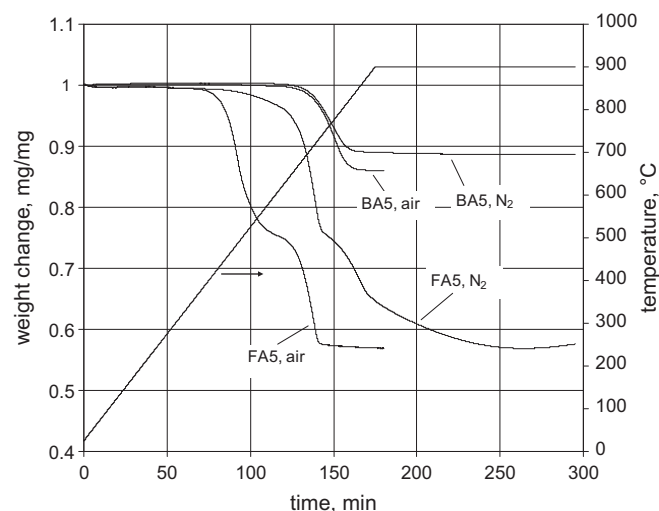
3.3. Free lime analysis

Results of the free lime analysis are presented in Table 11. The free lime content was typically well below 10% in all ashes except in BA6 and BA7, which showed 22% and 28%, respectively. These two samples were produced from combustion of petcoke at a higher bed temperature of ~950 °C, which causes the calcination of limestone even at the high CO₂ partial pressure of 0.08–0.09 MPa employed in this work, and hence a large amount of unreacted

Table 11

Free lime analysis results, wt.%.

Sample	Sample no.	Free Lime
Eastern bituminous + Havelock	BA1	5.2 ± 0.3
	FA1	0
	BA10	2.6 ± 0.8
	FA10	2.9 ± 0.5
Eastern bituminous + Czatkowice	BA11	8.2 ± 1.1
	FA11	1.3 ± 0.4
East Kentucky + Havelock	BA2	5.5 ± 0.5
	FA2T	6.7 ± 0.2
	FA2B	13.8 ± 0.9
	BA3	9.4 ± 1.3
	FA3	2.1 ± 0.1
	BA8	9.2 ± 1.0
	FA8	0.6 ± 0.5
East Kentucky + Czatkowice	BA9	10.7 ± 1.5
	FA9	1.6 ± 0.4
Coke + Havelock	BA4	10.1 ± 0.5
	FA4	5.0 ± 0.2
	BA5	4.7 ± 0.3
	FA5	0.9 ± 0.1
	RL5	1.1 ± 0.3
	BA6	22.0 ± 0.1
	FA6	4.2 ± 0.3
Coke + Czatkowice	BA7	28.2 ± 2.6
	FA7	1.4 ± 0.3
	BD7(850)	2.3 ± 0.4

**Fig. 2.** TGA analysis of BA5 and FA5 (coke + Havelock) in N₂ and air.

CaO was left in the ash. As the CFBC system was previously running at a lower temperature in oxy-fuel combustion mode, some CaCO_3 was also detected by XRD (as shown in Table 10). It should be noted that the free lime content in an air-fired CFBC ash is typically in the range of 20–30% [31]. Since free lime can cause high-pH leachate and other issues in ash disposal sites [38], oxy-fuel CFBC ash should perform much better in landfills because of its low free lime content. Calcite (CaCO_3) rather than free lime (CaO) in ash may also reduce the potential of agglomeration and fouling in CFBC due to chemical reaction sintering [35].

3.4. TGA analysis

The ash samples collected in this work were pyrolyzed in a Mettler–Toledo TGA at up to 900 °C, in either an inert (N_2) or an air environment, to determine the char carbon and CaCO_3 content based on their different decomposition temperature ranges in

pyrolysis. Char carbon starts to burn at 400 °C in air, while decomposition of CaCO_3 in either N_2 or air environment starts at ~650 °C. Char carbon and CaSO_4 can react at a higher temperature (~880 °C) in an inert environment to release SO_2 , which is typically seen in the fly ash samples that have high carbon content and the weight loss is shown on the TG curves. The flow rate of N_2 or air was 30 mL/min and the heating rate was 5 °C/min in all tests. As an example, Fig. 2 depicts typical TG curves of ashes produced from combustion of coke with addition of HL limestone (BA5 and FA5).

For BA5 CaCO_3 started to decompose at ~640 °C in N_2 . In air environment, the unburned carbon (slight amount) was burned off first at ~440 °C and decomposition of CaCO_3 then followed (~650 °C). For FA5 CaCO_3 started to decompose at a lower temperature of ~610 °C in N_2 . During pyrolysis in air, given that FA5 has much more char carbon, there was a significant weight loss shown on the TG curve starting at ~400 °C and decomposition of CaCO_3 followed at ~620 °C.

There were significant amounts of char carbon (>20%) detected in fly ash. When it was pyrolyzed in N_2 , a continuous and gradual weight loss was apparently seen on the TG curve, beginning at a temperature of ~880 °C. This process was found to last for about 1 h or longer. Similar results have been observed by Wu et al. [32] in a previous study on reactivation of fly ashes from industrial-scale CFBC boilers. Evidently, char carbon reacts with CaSO_4 , as there is sufficient char carbon and CaSO_4 together in the fly ash samples.

The char carbon and CaCO_3 contents determined by TGA are presented in Tables 5–7 and Table 12, respectively. As a crosscheck, the CaCO_3 content determined by TGA was in reasonable agreement with the XRD results shown in Tables 8–10, with slightly higher results found in the TGA analysis.

3.5. Surface analysis

Table 13 presents the surface analysis results of the ash samples produced from oxy-fuel combustion with different combinations of fuels and limestones. The BET surface area in the bed ash was

Table 12
CaCO₃ content determined by TGA, wt.%.

Fuel + limestone	Sample no.	CaCO ₃
Eastern bituminous + Havelock	BA(EB + HL)	9.4 ± 1.0
	FA(EB + HL)	13.0 ± 2.4
Eastern bituminous + Czatkowice	BA11	18.0 ± 1.4
	FA11	10.8 ± 7.4
East Kentucky + Havelock	BA(K + HL)	9.2 ± 12.8
	FA(K + HL)	17.8 ± 2.2
	FA2T	20.0 ± 4.2
	FA2B	15.0 ± 2.8
East Kentucky + Czatkowice	BA9	19.5 ± 4.9
	FA9	9.9 ± 3.0
Coke + Havelock	BA(C + HL)	17.2 ± 22.6
	FA(C + HL)	29.0 ± 12.4
	RL5	26.0 ± 5.7
Coke + Czatkowice	BA7	9.0 ± 1.2
	FA7	17.0 ± 8.5
	BD7(850)	49.0 ± 2.8

Table 13
Surface property analysis.

Fuel + limestone	Sample no.	BET surface area, m ² /g	Pore volume, cm ³ /g	Pore radius, nm
Eastern Bituminous + Havelock	BA1	0.78 ± 0.01	0.0026	6.66
	FA1	66.69 ± 0.80	0.040	1.20
	BA10	0.464 ± 0.0024	0.0012	5.21
	FA10	47.18 ± 0.618	0.0291	1.23
Eastern Bituminous + Czatkowice	BA11	1.25 ± 0.01	0.0019	3.03
	FA11	63.93 ± 0.85	0.0379	1.18
East Kentucky + Havelock	BA2	1.22 ± 0.0022	0.0042	6.87
	FA2T	27.97 ± 0.31	0.0201	1.44
	FA2B	7.93 ± 0.08	0.0074	1.87
	BA3	0.95 ± 0.0039	0.0018	3.79
	FA3	24.88 ± 0.24	0.018	1.47
	BA8	0.675 ± 0.0058	0.0012	3.59
	FA8	6.089 ± 0.0317	0.0089	2.93
	FA8	6.089 ± 0.0317	0.0089	2.93
East Kentucky + Czatkowice	BA9	1.003 ± 0.0051	0.0022	4.35
	FA9	26.09 ± 0.28	0.0198	1.51
Coke + Havelock	BA4	0.51 ± 0.0034	0.00107	4.16
	FA4	20.23 ± 0.26	0.0145	1.44
	BA5	0.64 ± 0.0095	0.0011	3.42
	FA5	8.72 ± 0.070	0.0083	1.91
	RL5	0.49 ± 0.0032	0.0011	4.63
	BA6	0.758 ± 0.0027	0.00167	4.42
	FA6	5.743 ± 0.031	0.0067	2.34
	FA6	5.743 ± 0.031	0.0067	2.34
Coke + Czatkowice	BA7	0.91 ± 0.0027	0.0015	3.30
	FA7	4.87 ± 0.029	0.0068	2.78
	BD7(850)	0.52 ± 0.0034	0.0016	6.06

between 0.5 and 1.2 m²/g, regardless of the fuel/limestone combination used. The surface area of the BA was within the surface area range of bed ash from air-firing CFBC [39]. The BET surface area in the FA was much greater than that in the BA, in the range of 5–67 m²/g. The FA produced from combustion of EB coal had a very large BET area, which was about 50–80 times that of the BA. The surface area of the FA obtained from this unit was also larger than the surface area normally seen in air-firing industrial-scale CFBC fly ash (5–10 m²/g) [32]. One of the possible explanations for the larger surface area observed in this study is the effect of the high carbon content remaining in the FA. However, it should also be noted that most of the fly ash produced from coke firing had much smaller BET areas compared to those from coal firing, although they had comparable amounts of carbon in the FA. More investigation is, therefore, needed to clarify the causes of the larger surface area. The fly ash collected from the top of the baghouse ash bin (FA2T) had a larger surface area than the ash from the bottom portion (FA2B). In most cases, the pore volume in the FA was 10 times larger than that in the BA. The pore radius in the FA, in contrast, was much smaller. This implies that FA should have a much larger amount of small pores than BA.

4. Conclusions

Oxy-fuel combustion experiments were successfully conducted on a modified 100 kW_{th} mini-CFBC unit at CanmetENERGY. The char carbon content was much higher in the fly ash (>20%) than in the bed ash. The substantial amount of char carbon in the fly ash was believed due mainly to the short residence time available for fine fuel particles to burn out in the small-scale unit used in this work. When fly ash was pyrolyzed in a TGA in a N₂ environment, it was shown that char carbon reacted with CaSO₄ at temperatures over 880 °C. The CaCO₃ content was higher in the ashes from oxy-fuel combustion in this unit compared to conventional air-firing CFBC ashes in which the CaCO₃ content was typically undetectable. This confirms that sulfur capture under oxy-fuel combustion conditions is *via* direct sulfation mode (up to the equilibrium temperature corresponding to the CO₂ partial pressures in the combustor), due to high partial pressure of CO₂, which is similar to the environment found in a PFBC and prevents calcination of limestone particles. Limestone utilization for oxy-fuel combustion in this unit was found to be lower than that in commercial-scale air-firing CFBC. However, better limestone performance resulted during combustion of petcoke at a slightly higher bed temperature of ~950 °C (indirect sulfation occurred at this temperature), which was in good agreement with previous observations. The BET surface area and the pore volume in the fly ash were much larger than those of the bed ash. There are many more smaller-sized pores existing in the fly ash. The free lime content in either bed or fly ash produced from this unit was much lower than that in air-firing CFBC ash samples. Since the dominant calcium compounds in oxy-fuel CFBC ash were CaCO₃ and CaSO₄ rather than CaO (at least for tests carried out at temperatures below 900 °C), disposal behavior of oxy-fuel CFBC ash should be preferable to that of air-firing CFBC ashes. Issues related to high-pH leachate and landfill cracking may be expected to be reduced. Agglomeration and fouling due to continued sulfation of CaO in CFBCs firing high-sulfur, low-ash fuels (such as petcoke) may also be mitigated.

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